



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

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SECP3623 DATABASE PROGRAMMING

SECTION 02

Assignment 2

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1.0 Start MongoDB Shell

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.6.0
Microsoft Windows [Version 10.0.22631.6199]
(c) Microsoft Corporation. All rights reserved.

C:\Users\LENOVO>mongosh
Current Mongosh Log ID: 696c9650c31418088e628c9f
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.6.0
Using MongoDB:      8.2.3
Using Mongosh:      2.6.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
  The server generated these startup warnings when booting
  2026-01-17T12:48:48.135+08:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> |
```

A.1.1 Create / Switch Database

```
test> use clinicdb
switched to db clinicdb
clinicdb> |
```

A.1.2 Create collections: appointments & loginSessions

```
clinicdb> db.createCollection("appointments")
{ ok: 1 }
clinicdb> |

clinicdb> db.createCollection("loginSessions")
{ ok: 1 }
clinicdb> |
```

A.1.3 Verify Collections

```
clinicdb> show collections
appointments
loginSessions
clinicdb> |
```

A.2.1 Insert Data into appointments

```
clinicdb> db.appointments.insertMany([
... {
...   appointmentId: "A001",
...   patientName: "Sarah Khalila",
...   doctor: "Dr Han",
...   department: "Cardiology",
...   status: "COMPLETED",
...   fee: 250,
...   appointmentDate: new Date("2026-01-10")
... },
... {
...   appointmentId: "A002",
...   patientName: "Nur Jamila",
...   doctor: "Dr Iskandar",
...   department: "Dermatology",
...   status: "PENDING",
...   fee: 120,
...   appointmentDate: new Date("2026-01-12")
... },
... ])
```

```
... {
...   appointmentId: "A003",
...   patientName: "Danish Uqael",
...   doctor: "Dr Ariana",
...   department: "Neurology",
...   status: "COMPLETED",
...   fee: 300,
...   appointmentDate: new Date("2026-01-15")
... },
... {
...   appointmentId: "A004",
...   patientName: "Yusuf Iskandar",
...   doctor: "Dr Redza",
...   department: "Pediatrics",
...   status: "CANCELLED",
...   fee: 80,
...   appointmentDate: new Date("2026-01-18")
... },
... {
...   appointmentId: "A005",
...   patientName: "Mohd Eidlan",
...   doctor: "Dr Mawar",
...   department: "Cardiology",
...   status: "CANCELLED",
...   fee: 200,
...   appointmentDate: new Date("2026-01-20")
... },
... ])
```

```
... {
...   appointmentId: "A006",
...   patientName: "Nur Eleisha",
...   doctor: "Dr Farhat",
...   department: "Gynecology",
...   status: "PENDING",
...   fee: 150,
...   appointmentDate: new Date("2026-01-22")
... }
... ])
```

RESULT A.2.1:

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('696cb0af3da75550e8628ca0'),
    '1': ObjectId('696cb0af3da75550e8628ca1'),
    '2': ObjectId('696cb0af3da75550e8628ca2'),
    '3': ObjectId('696cb0af3da75550e8628ca3'),
    '4': ObjectId('696cb0af3da75550e8628ca4'),
    '5': ObjectId('696cb0af3da75550e8628ca5')
  }
}
```

A.2.2 Insert Data into loginSessions

```
clinedb> db.loginSessions.insertMany([
... {
...   staffId: "S001",
...   staffName: "Maria",
...   role: "Nurse",
...   status: "ACTIVE",
...   ipAddress: "192.168.1.10",
...   loginTime: new Date("2026-01-20T08:30:00")
... },
... {
...   staffId: "S002",
...   staffName: "Aisyah",
...   role: "Admin",
...   status: "INACTIVE",
...   ipAddress: "192.168.1.11",
...   loginTime: new Date("2026-01-20T09:00:00")
... },
... {
...   staffId: "S003",
...   staffName: "Harlene",
...   role: "Receptionist",
...   status: "ACTIVE",
...   ipAddress: "192.168.1.12",
...   loginTime: new Date("2026-01-20T09:30:00")
... },
... ])
```

```
...   staffId: "S004",
...   staffName: "Farhat",
...   role: "Doctor",
...   status: "ACTIVE",
...   ipAddress: "192.168.1.13",
...   loginTime: new Date("2026-01-20T10:00:00")
... },
... {
...   staffId: "S005",
...   staffName: "Redza",
...   role: "Doctor",
...   status: "ACTIVE",
...   ipAddress: "192.168.1.14",
...   loginTime: new Date("2026-01-20T10:30:00")
... },
... {
...   staffId: "S006",
...   staffName: "Daisy",
...   role: "Nurse",
...   status: "ACTIVE",
...   ipAddress: "192.168.1.15",
...   loginTime: new Date("2026-01-20T11:00:00")
... }
... ])
{
```

RESULT A.2.2:

```
acknowledged: true,
insertedIds: {
  '0': ObjectId('696cb69b0680a54fb0628ca0'),
  '1': ObjectId('696cb69b0680a54fb0628ca1'),
  '2': ObjectId('696cb69b0680a54fb0628ca2'),
  '3': ObjectId('696cb69b0680a54fb0628ca3'),
  '4': ObjectId('696cb69b0680a54fb0628ca4'),
  '5': ObjectId('696cb69b0680a54fb0628ca5')
}
```

B.1 Advanced Query & Sorting

```
clinicdb> db.appointments.find(
... {
... status: "COMPLETED",
... fee: { $gte: 100, $lte: 300 }
... },
... {
... _id: 0,
... appointmentId: 1,
... patientName: 1,
... department: 1,
... fee: 1,
... }
... ).sort({department: 1, fee: -1})
[
```

RESULT B.1:

```
{
  appointmentId: 'A001',
  patientName: 'Sarah Khalila',
  department: 'Cardiology',
  fee: 250
},
{
  appointmentId: 'A003',
  patientName: 'Danish Uqael',
  department: 'Neurology',
  fee: 300
}
]
```

B.2 Aggregation Summary

```
clinicdb> db.appointments.aggregate([
... {
...   $group: {
...     _id: "$department",
...     totalAppointments: { $sum: 1 },
...     totalFees: { $sum: "$fee" }
...   }
... },
... {
...   $sort: { totalFees: -1 }
... }
... ])
```

RESULT B.2:

```
[
  { _id: 'Cardiology', totalAppointments: 2, totalFees: 450 },
  { _id: 'Neurology', totalAppointments: 1, totalFees: 300 },
  { _id: 'Gynecology', totalAppointments: 1, totalFees: 150 },
  { _id: 'Dermatology', totalAppointments: 1, totalFees: 120 },
  { _id: 'Pediatrics', totalAppointments: 1, totalFees: 80 }
]
```

C.1 Active Staff Sessions

```
clinicdb> db.loginSessions.find(
... {status: "ACTIVE" },
... {_id: 0, staffName: 1, role: 1, ipAddress: 1, loginTime: 1 }
... ).sort({ loginTime: -1})
```

RESULT C.1

```
[
  {
    staffName: 'Daisy',
    role: 'Nurse',
    ipAddress: '192.168.1.15',
    loginTime: ISODate('2026-01-20T03:00:00.000Z')
  },
  {
    staffName: 'Redza',
    role: 'Doctor',
    ipAddress: '192.168.1.14',
    loginTime: ISODate('2026-01-20T02:30:00.000Z')
  },
  {
    staffName: 'Farhat',
    role: 'Doctor',
    ipAddress: '192.168.1.13',
    loginTime: ISODate('2026-01-20T02:00:00.000Z')
  },
  {
    staffName: 'Harlene',
    role: 'Receptionist',
    ipAddress: '192.168.1.12',
    loginTime: ISODate('2026-01-20T01:30:00.000Z')
  },
  {
    staffName: 'Maria',
    role: 'Nurse',
    ipAddress: '192.168.1.10',
    loginTime: ISODate('2026-01-20T00:30:00.000Z')
  }
]
```

C.2 Distinct & Count

C.2.1 Distinct IP addresses of ACTIVE staff

```
clinicdb> db.loginSessions.distinct("ipAddress", { status: "ACTIVE" })
[
  '192.168.1.10',
  '192.168.1.12',
  '192.168.1.13',
  '192.168.1.14',
  '192.168.1.15'
]
```

C.2.2 Calculate the number of unique staff members who currently have ACTIVE sessions.

```
clinicdb> db.loginSessions.distinct("staffId", { status: "ACTIVE" }).length
5
```

D.1 Compound Index

```
clinicdb> db.appointments.createIndex({ department: 1, status: 1})  
department_1_status_1
```

D.2 Update & Delete Operations

```
clinicdb> db.appointments.updateMany(  
... {status: "CANCELLED"},  
... {$set: {status: "ARCHIVED"}}  
... )  
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 2,  
  modifiedCount: 2,  
  upsertedCount: 0  
}
```

```
clinicdb> db.loginSessions.deleteMany({ status: "INACTIVE" })  
{ acknowledged: true, deletedCount: 1 }
```